

Distributed Rooftop Solar Development in Bangladesh Commercial and Industrial Facilities: Microeconomic Analysis and Financing Structures for the Netso Energy Reference Project

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Abstract

Distributed commercial and industrial (C&I) rooftop solar in Bangladesh addresses grid instability and volatile commercial utility tariffs under severe land-scarcity constraints. This paper presents the empirical and microeconomic framework of Netso Energy’s reference project: an 80 kWp commercial photovoltaic installation at the Chittagong Grammar School (CGS) campus in Chattogram, Bangladesh. We evaluate operational generation physics, levelized tariff arbitrage against the Bangladesh Power Development Board (BPDB) Medium Tension category 2 (MT-2) rate, and project financing mechanisms. Operating at a 16.5% capacity factor, the 80 kWp facility yields 115,632 kWh annually, offsetting 98.0% of site demand. Under the baseline power purchase agreement (PPA) of BDT 10.00/kWh versus the true variable utility rate of BDT 12.98/kWh, the client achieves a 23.0% variable expenditure savings. Under an 80/20 Infrastructure Development Company Limited (IDCOL) concessional debt structure (6.0% p.a., 10-year amortization), the asset generates a Year-1 Debt Service Coverage Ratio (DSCR) of 2.25 $\times$, an equity payback period of 1.47 years, and a 20-year levered equity internal rate of return (IRR) of 68.7%. We examine the policy sensitivity of the FY2026–27 zero-tariff import schedule and evaluate an alternative non-dilutive 35% gross revenue-share investment structure underwritten by Beringia Energy.

1 Introduction

Industrial and commercial power consumers across Bangladesh encounter rising tariff schedules, load shedding, and grid transmission congestion [3]. While conventional utility-scale solar farms require substantial contiguous acreage that is scarce in urbanized and agricultural regions, commercial concrete rooftops provide an unmonetized structural layer for distributed electricity generation [3].

Netso Energy operates as an independent developer deploying build-own-operate (BOO) and renewable energy service company (RESCO) infrastructure. The foundational physical hardware platform comprises the Solar Pergola, an engineered modular structural canopy that provides simultaneous shading, weather shelter, and photovoltaic power generation on flat rooftops [3].

To establish bankability and investor underwriting standards, Netso Energy structured an 80 kWp reference deployment at the Chittagong Grammar School (CGS) facility in Chattogram [1, 4]. This paper presents the end-to-end techno-economic specification, empirical utility audit analysis, financial capitalization models, and contract risk mitigations governing this reference installation.

*Netso Energy. Ground-truth constants generated via `System/scripts/core_economics.py`.

2 Methods

2.1 Physical Generation Model

Solar photovoltaic generation is modeled using site-specific empirical solar irradiance parameters for Chattogram, Bangladesh. Annual electrical energy generation E_1 in kilowatt-hours (kWh) during Year 1 is governed by:

$$E_1 = P_{dc} \times 8760 \times CF \quad (1)$$

where $P_{dc} = 80$ kWp is the direct-current nameplate capacity, 8760 represents total annual operating hours, and $CF = 0.165$ (16.5%) is the empirical capacity factor for Chattogram [1].

Annual energy yield degrades continuously across operating lifespan $t \in [1, 20]$ at an annual degradation coefficient $\delta = 0.005$ (0.5%/year):

$$E_t = E_1 \times (1 - \delta)^{t-1} \quad (2)$$

2.2 Utility Tariff Decomposition and Tariff Arbitrage

Client historical billing data is audited from 12 consecutive months of MT-2 utility statements. Total annual expenditure C_{grid} decomposes into fixed demand charges F_{demand} and variable energy charges:

$$C_{grid} = F_{demand} + (Q_{grid} \times R_{var}) \quad (3)$$

where $Q_{grid} = 118,000$ kWh/yr, $C_{grid} = \text{BDT } 1,748,056/\text{yr}$, and $F_{demand} = \text{BDT } 216,000/\text{yr}$. The true variable rate R_{var} is:

$$R_{var} = \frac{C_{grid} - F_{demand}}{Q_{grid}} = \frac{1,748,056 - 216,000}{118,000} = \text{BDT } 12.98/\text{kWh} \quad (4)$$

Under the Netso PPA, solar energy is delivered at contract rate $R_{ppa} = \text{BDT } 10.00/\text{kWh}$. The client's true percentage savings margin S_{margin} and Year-1 financial savings ΔC_{client} are:

$$S_{margin} = \frac{R_{var} - R_{ppa}}{R_{var}} = \frac{12.98 - 10.00}{12.98} = 23.0\% \quad (5)$$

$$\Delta C_{client} = E_1 \times (R_{var} - R_{ppa}) = 115,632 \times (12.98 - 10.00) = \text{BDT } 344,991 \quad (6)$$

2.3 Project Capitalization and Debt Service Mechanics

Capital expenditure (CAPEX) per kilowatt K is modeled across two distinct cost environments:

- **Scenario A (Baseline):** $K_A = \text{BDT } 55,000/\text{kW}$, EPC hard cost of $\text{BDT } 40,000/\text{kW}$ plus developer margin of $\text{BDT } 15,000/\text{kW}$. Total CAPEX: $I_A = \text{BDT } 4,400,000$ [1].
- **Scenario B (Policy Upside):** The FY2026–27 National Budget eliminates import duties on PV modules, inverters, and racking. $K_B = \text{BDT } 40,000/\text{kW}$ ($I_B = \text{BDT } 3,200,000$) [1].

Financing is modeled via concessional IDCOL facility parameters: debt ratio $d = 0.80$ (80%), equity ratio $e = 0.20$ (20%), interest rate $r = 0.06$ (6.0% p.a.), and tenor $n = 10$ years. Annual debt service A_{debt} is calculated via standard annuity amortization:

$$A_{debt} = (d \cdot I) \times \frac{r(1+r)^n}{(1+r)^n - 1} \quad (7)$$

Operating expenditure (OPEX) is set to $O_1 = \text{BDT } 1,000/\text{kW}$ ($\text{BDT } 80,000/\text{year}$), escalating at $g_{opex} = 0.025$ (2.5%/year). Gross revenue R_t escalates at $g_{ppa} = 0.03$ every 3 years

(3% triennial). The Debt Service Coverage Ratio in year t ($DSCR_t$) and Free Cash Flow to Equity ($FCFE_t$) are:

$$DSCR_t = \frac{EBITDA_t}{A_{\text{debt}}} = \frac{R_t - O_t}{A_{\text{debt}}} \quad (8)$$

$$FCFE_t = EBITDA_t - A_{\text{debt}} \quad (9)$$

2.4 Alternative Capital Structure: Pure Revenue-Share Model

As an alternative to bilateral debt facilities, a pure revenue-share investment structure was underwritten with Beringia Energy [5, 6]. Beringia injects 100% of project CAPEX ($I_A = \text{BDT } 4,400,000$) in three tranching drawdowns: 30% at contract execution (BDT 1.32M), 50% at EPC hardware mobilization (BDT 2.20M), and 20% at Commercial Operation Date (BDT 0.88M). In return, Beringia receives a 35% allocation of gross PPA revenues until achieving a cumulative 15.0% unlevered IRR ($T_{\text{exit}} \approx 17.6$ years), upon which all asset cash flows revert 100% to Netso Energy [6].

3 Results

3.1 Baseline Energy and Techno-Economic Metrics

Application of Equation (1) to the 80 kWp array yields an annual output of $E_1 = 115,632$ kWh. Against the CGS audited annual consumption of 118,000 kWh, the installation provides an on-site energy coverage ratio of 98.0% [1]. Table 1 summarizes the physical, operational, and capital metrics across baseline and policy upside scenarios.

Table 1: Techno-Economic and Capitalization Comparison (80 kWp System)

Parameter / Metric	Scenario A (Baseline)	Scenario B (Policy Upside)	Delta (Δ)
Nameplate Capacity	80 kWp	80 kWp	0.0%
Annual Generation (Year 1)	115,632 kWh	115,632 kWh	0.0%
Capacity Factor (Chattogram)	16.5%	16.5%	0.0%
Annual Degradation Rate	0.5% p.a.	0.5% p.a.	0.0%
CAPEX per kWp	BDT 55,000	BDT 40,000	-27.3%
Total Project Cost (I)	BDT 4,400,000	BDT 3,200,000	-27.3%
Equity Contribution (20%)	BDT 880,000	BDT 640,000	-27.3%
IDCOL Debt Principal (80%)	BDT 3,520,000	BDT 2,560,000	-27.3%
Gross Revenue (Year 1)	BDT 1,156,320	BDT 1,156,320	0.0%
OPEX (Year 1)	BDT 80,000	BDT 80,000	0.0%
EBITDA (Year 1)	BDT 1,076,320	BDT 1,076,320	0.0%
Annual Debt Service (A_{debt})	BDT 478,255	BDT 347,822	-27.3%
Year-1 FCFE	BDT 598,065	BDT 728,498	+21.8%
Debt Service Coverage Ratio (DSCR)	2.25×	3.09×	+37.3%
Project Payback Period	4.1 years	3.0 years	-26.8%
Equity Payback Period	1.47 years	0.88 years	-40.1%
20-Year Levered Equity IRR	68.7%	114.1%	+66.1%
10-Year Cumulative FCFE	BDT 6,140,663	BDT 7,444,995	+21.2%

3.2 Cash Flow and Revenue Dynamics

Figure 1 depicts the projected 10-year cash flow waterfall under Scenario A, illustrating gross revenues, operating costs, debt service obligations, and net FCFE distributions.

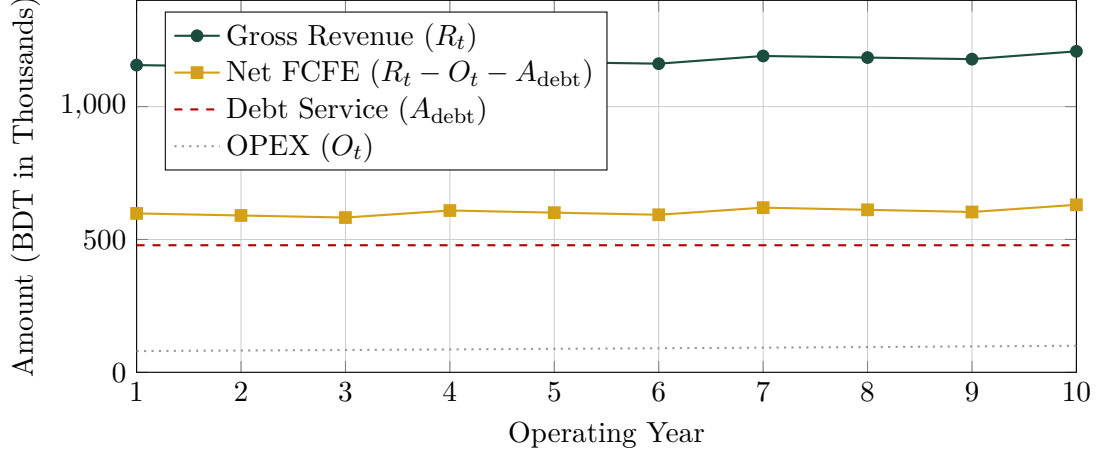


Figure 1: Ten-year financial cash flow projection for the Netso Energy 80 kWp CGS reference project under baseline debt terms (Scenario A), demonstrating EBITDA stability and debt coverage.

3.3 Beringia Revenue-Share Stress Testing

Under the Beringia Energy pure revenue-share structure, Year-1 gross revenue of BDT 1,156,320 yields BDT 374,712 (35%) to the capital provider and BDT 781,608 (65%) to Netso Energy. Net of 100% absorbed OPEX (BDT 80,000), Netso retains BDT 701,608 in Year-1 operating profit [6].

Table 2 evaluates project sensitivity across operational and credit shock scenarios [7].

Table 2: Beringia Revenue-Share Structure Sensitivity and Downside Stress Test

Scenario	Mechanism & Impact	Netso Profit (Yr 1)	Investor Return Impact
Baseline	35% gross revenue split; 15.0% unlevered IRR hurdle	BDT 701,608	Payback $\approx$ 17.6 years
Revenue -10%	Generation deficit or tariff floor (BDT 9.00/kWh)	BDT 621,608 (-11.4%)	Payback expands to 19.6 yrs
OPEX +50%	OPEX increases to BDT 120,000/yr (Netso absorbed)	BDT 661,608 (-5.7%)	Unaffected (Gross split)
Year 5 Buyback	Early buyout exercised at 1.2 $\times$ remaining unrecovered value	N/A	Target exit at BDT 5.67M
Default Year 3	CGS payment default; 6-mo bank guarantee + escrow drawn	Asset retained	Net loss capped at BDT 546k

4 Discussion

4.1 Structural Advantages of the True Variable Rate Method

Conventional solar developers in Bangladesh frequently calculate savings estimates against blended electricity rates ($R_{\text{blend}} \approx$ BDT 14.81/kWh), which incorrectly incorporate fixed max-

imum demand charges. Because fixed demand charges persist irrespective of distributed generation, projecting savings against blended figures overstates customer economic benefits. Netso Energy’s methodology isolates the pure variable component ($R_{\text{var}} = \text{BDT } 12.98/\text{kWh}$), establishing an auditable 23.0% cost reduction that withstands institutional scrutiny [1, 9].

4.2 Policy Upside and Capital Optimization

The statutory exemptions in the FY2026–27 National Budget reducing clean energy component import tariffs to 0% provide substantial upside for distributed infrastructure [1]. Under Scenario B, CAPEX declines from BDT 55,000/kW to BDT 40,000/kW. Rather than adjusting PPA tariffs downward, retaining the contract rate at BDT 10.00/kWh increases project DSCR from $2.25\times$ to $3.09\times$ and elevates levered equity IRR from 68.7% to 114.1%, enabling rapid capital recycling across pipeline facilities [1].

4.3 Contractual Governance and Risk Mitigation

In the Beringia revenue-share model, counterparty and structural risks are mitigated through four primary covenants:

1. **Conditions Precedent:** Execution requires grid sanction load verification (≥ 80 kW under the updated SREDA Net Metering Guidelines 2025 allowing up to 100% of sanctioned load, improved from the previous 70% ceiling that required ≥ 115 kW) and Chattogram Development Authority (CDA) rooftop structural engineering certification [6].
2. **Payment Security:** Direct PPA billing flows into an irrevocable escrow account with an active 6-month first-demand bank guarantee (BDT 578,000) [6].
3. **Downside Preservation:** In the event of client default, modular pergola infrastructure can be unbolted and redeployed to alternative commercial properties [7].
4. **Arbitration Seat:** Legal disputes are bound exclusively to the Bangladesh Arbitration Act 2001 with legal seat in Dhaka [6].

5 Conclusion

This study validates the commercial, physical, and financial model of Netso Energy’s 80 kWp reference solar pergola installation at CGS Chattogram. By coupling modular rooftop structural engineering with conservative variable tariff arbitrage (BDT 10.00/kWh PPA vs. BDT 12.98/kWh grid variable), the project delivers 23.0% recurring savings to the client while producing robust debt service coverage ($2.25\times$) and high levered equity returns (68.7% IRR). Non-dilutive revenue-share instruments offer an effective institutional bridge for early portfolio scale prior to long-term concessional debt refinancing.

References

References

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